

Experiment 1

1. Aim

To design and simulate a **Full Adder circuit** using basic logic gates (AND, OR, XOR) in **Logisim Evolution**, verify its truth table, and observe the Sum and Carry outputs for all possible combinations of three binary inputs.

2. Theory

A **Full Adder** is a combinational logic circuit that performs the addition of three single-bit binary numbers: two operand bits (A, B) and a carry-in bit (C_{in}). It produces two outputs: a **Sum** bit and a **Carry-out** (C_{out}) bit.

The Boolean expressions governing the Full Adder are:

- **Sum** = $A \oplus B \oplus C_{in}$
- **C_{out}** = $(A \cdot B) + (B \cdot C_{in}) + (A \cdot C_{in})$

A Full Adder can be constructed using two Half Adders and an OR gate, or directly using three XOR, three AND, and one OR gate. It forms the basic building block of all binary arithmetic units.

Truth Table:

A	B	C_{in}	Sum	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

3. Procedure

1. Open **Logisim Evolution** and create a new project file.
2. Add three input pins to the canvas and label them **A**, **B**, and **C_{in}** .
3. Place two **XOR gates** on the canvas. Connect A and B to the first XOR gate to obtain the intermediate result $A \oplus B$.
4. Connect the output of the first XOR gate and C_{in} to the second XOR gate. The output of this gate represents the **Sum**.

5. Place three **AND gates**. Connect (A, B), (B, C_{in}), and (A, C_{in}) to each AND gate respectively.
6. Connect the outputs of all three AND gates to an **OR gate**. The output represents C_{out} .
7. Add two output pins and label them **Sum** and **Carry**. Wire them to the respective gate outputs.
8. Use the **Poke tool** to toggle all eight input combinations (000 through 111) and verify the outputs against the expected truth table.
9. Take a screenshot of the completed and verified circuit.

4. Circuit Design / Implementation

The Full Adder circuit was constructed in **Logisim Evolution** using the following gate configuration:

- **2 × XOR gates** — to compute the Sum output.
- **3 × AND gates** — to generate partial carry products.
- **1 × OR gate** — to combine partial carries into C_{out} .

Input pins A, B, and C_{in} were routed to the appropriate gate inputs using Logisim's wiring tool. The outputs (Sum and Carry) were connected to output pins and a hex display for real-time visual verification. All wires were colour-coded green to indicate active logic-high states during simulation.

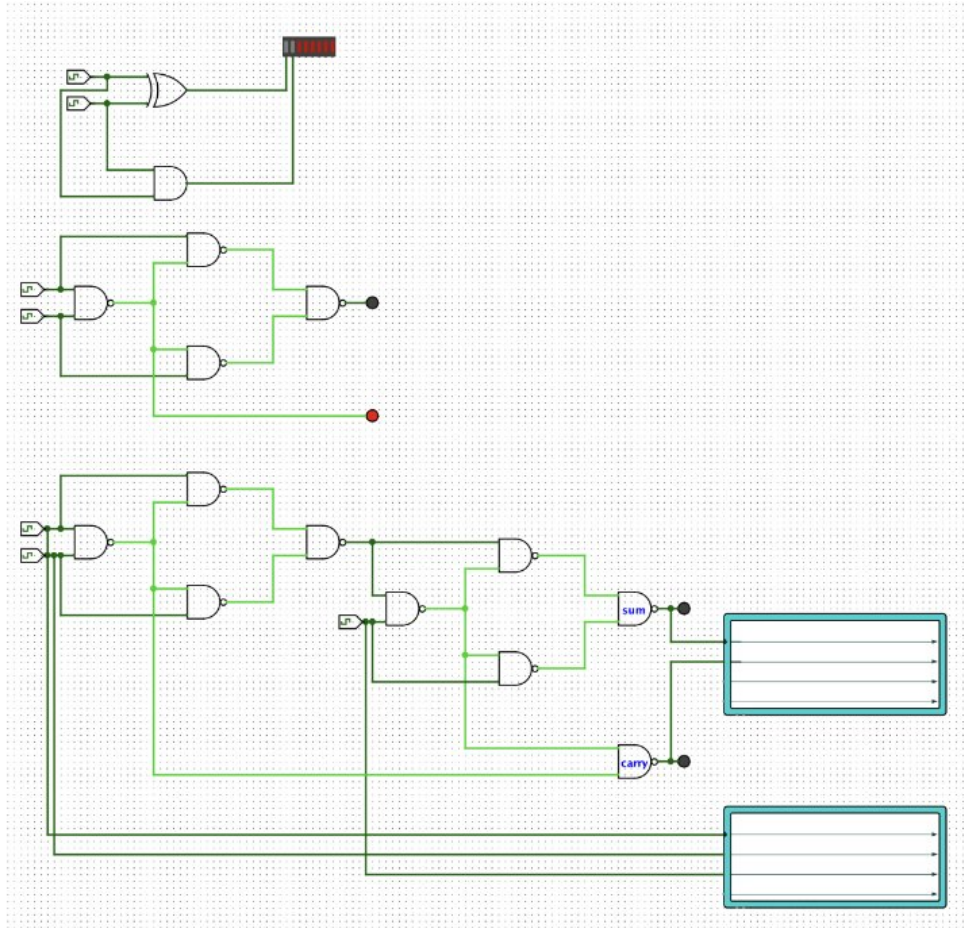


Figure 1: Full Adder circuit implemented in Logisim Evolution showing inputs A, B, C_{in} and outputs Sum, Carry.

5. Result

The Full Adder circuit was successfully designed and simulated in Logisim Evolution. All eight possible input combinations (000 to 111) were tested, and the observed Sum and Carry outputs matched the expected values in the truth table. The circuit correctly computed binary addition with carry for every test case.

6. Conclusion

Through this experiment, the working principle and gate-level implementation of a **Full Adder** were understood and verified. Logisim Evolution provided an effective simulation environment to validate the Boolean logic before physical implementation. The experiment demonstrated how combinational circuits form the foundation of binary arithmetic operations in digital systems.